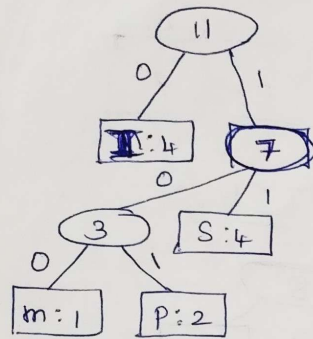


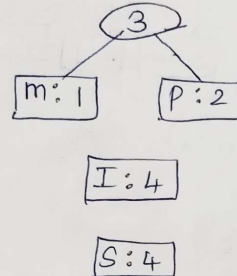
Huffman coding

M=1, S=4, P=2, I=4

word $\Rightarrow$ Mississippi



Character	Freq
M	1
I	4
P	2
S	4



M $\Rightarrow$ 100

P $\Rightarrow$ 101

S $\Rightarrow$ 11

I $\Rightarrow$ 0

Direct - Fixed reps $\Rightarrow 11 \times 8 \Rightarrow 88$ bits

$\Rightarrow 88 - 64$

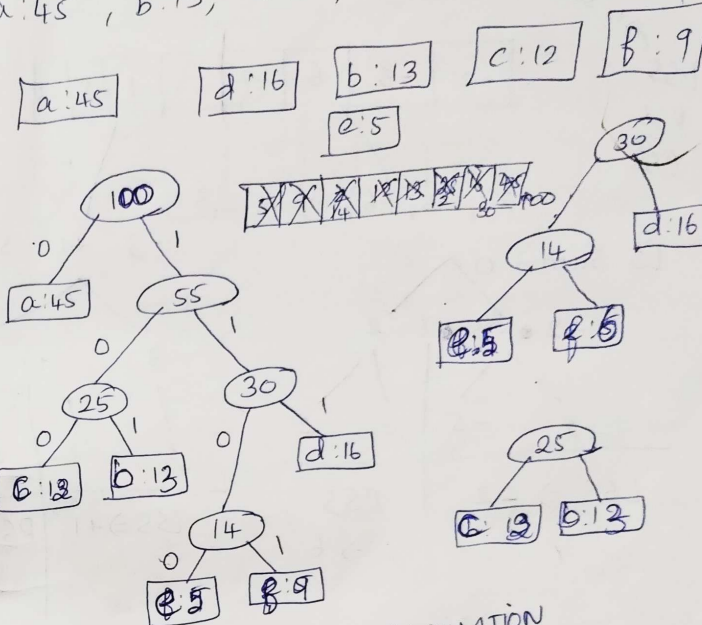
$\Rightarrow 24$ bits Saved

BIT CALCULATION

Characters	Frequency	No of bits	Calculation Freq x no of bits
M	1	100	$1 \times 3 \Rightarrow 3$
I	4	0	$4 \times 1 \Rightarrow 4$
P	2	101	$2 \times 3 \Rightarrow 6$
S	4	11	$4 \times 2 \Rightarrow 8$
Each character 8 bits $\Rightarrow 4 \times 8$ $\Rightarrow 32$ bits	11 bits	-	21 bits

total bits to represent characters $\Rightarrow 32 + 11 + 21 \Rightarrow 64$ bits

a:45, b:13, c:12, d:16, e:5, f:9



BIT CALCULATION

a = 0

b = 100

c = 100

d = 111

e = 1101

f = 1100

Characters	Freq	No of bits	Calculation Freq x No of bits
a	45	0	$45 \times 1 \Rightarrow 45$
b	13	100	$13 \times 3 \Rightarrow 39$
c	12	100	$12 \times 3 \Rightarrow 36$
d	16	111	$16 \times 3 \Rightarrow 48$
e	5	1100	$5 \times 4 \Rightarrow 20$
f	9	1100	$9 \times 4 \Rightarrow 36$

Each char
8 bits
 $6 \times 8 \Rightarrow 48$ bits

224

Total bits $\Rightarrow 48 + 100 + 224 \Rightarrow 372$ bits

Total bits needed for representation $\Rightarrow (100 \times 8)$
fixed $\Rightarrow 800$
 $\Rightarrow 800 - 372 \Rightarrow 428$ bits saved